CLIMATE CLASSIFICATION (KOPPEN) / INDIA CLIMATIC REGIONS

CLASSIFICATION DESIGN GATE → KOPPEN MAJOR GROUPS → TROPICAL CODE GRAMMAR → DRY-CLIMATE GATE → C-D THRESHOLD FIREWALL → NORMAL-TO-TREND LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 CLASSIFICATION DESIGN GATE

CLASSIFICATION DESIGN GATE | COMPARISON, EXPLANATION OR PLANNING | TEMPERATURE, RAIN OR WATER BALANCE | THRESHOLDS, PERIOD, SCALE, INTERPOLATION | USEFUL MODEL WITH EXPLICIT LIMITS | MUST REMEMBER | KÖPPEN CLASSIFICATION USES TEMPERATURE AND PRECIPITATION THRESHOLDS WITH SEASONALITY

PURPOSE → comparison, explanation or planning

VARIABLES → temperature, rain or water balance → METHOD → thresholds, period, scale, interpolation → OUTPUT → useful model with explicit limits

CORE CLAIM
- PURPOSE → comparison, explanation or planning
- VARIABLES → temperature, rain or water balance

MECHANISM
- METHOD → thresholds, period, scale, interpolation
- OUTPUT → useful model with explicit limits

CONSEQUENCE / LIMIT
- MUST REMEMBER: Köppen classification uses temperature and precipitation thresholds with seasonality letters
- India must be mapped as regional climate regimes shaped by latitude, altitude, continentality, relief and monsoon circulation rather

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Köppen classification uses temperature and precipitation thresholds with seasonality letters; India must be mapped as regional climate regimes shaped by latitude, altitude, continentality, relief and monsoon circulation rather than memorised codes alone.

01

STAGE 01 KOPPEN MAJOR GROUPS

KOPPEN MAJOR GROUPS | TROPICAL THERMAL REGIME | DRYNESS THRESHOLD FIRST | COLDEST-MONTH CONVENTION MATTERS | WARMEST MONTH BELOW 10 C; H ADDED LATER

A → tropical thermal regime

B → dryness threshold first → C / D → coldest-month convention matters → E → warmest month below 10 C; H added later

CORE CLAIM
- A → tropical thermal regime
- B → dryness threshold first

MECHANISM
- C / D → coldest-month convention matters
- E → warmest month below 10 C; H added later

CONSEQUENCE / LIMIT
- A → tropical thermal regime
- B → dryness threshold first

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: E → warmest month below 10 C; H added later.

02

STAGE 02 TROPICAL CODE GRAMMAR

TROPICAL CODE GRAMMAR | NO DRY SEASON | MONSOON SHORT DRY SEASON LOGIC | DRY WINTER | DRY SUMMER | FIRST LETTER CLIMATE, SECOND RAINFALL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
Af → no dry season	Aw / As → dry winter / dry summer	Af → no dry season
Am → monsoon short dry season logic	RULE → first letter climate, second rainfall	Am → monsoon short dry season logic

CORE CLAIM
- Af → no dry season
- Am → monsoon short dry season logic

MECHANISM
- Aw / As → dry winter / dry summer
- RULE → first letter climate, second rainfall

CONSEQUENCE / LIMIT
- Af → no dry season
- Am → monsoon short dry season logic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Aw / As → dry winter / dry summer.

03

STAGE 03 DRY-CLIMATE GATE

Af → no dry season	Aw / As → dry winter / dry summer	Af → no dry season
Am → monsoon short dry season logic	RULE → first letter climate, second rainfall	Am → monsoon short dry season logic

CORE CLAIM
- Af → no dry season
- Am → monsoon short dry season logic

MECHANISM
- Aw / As → dry winter / dry summer
- RULE → first letter climate, second rainfall

CONSEQUENCE / LIMIT
- Af → no dry season
- Am → monsoon short dry season logic

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Aw / As → dry winter / dry summer.

03

STAGE 03 DRY-CLIMATE GATE

DRY-CLIMATE GATE · TEMPERATURE + RAIN SEASON · DRYNESS THRESHOLD · PRECIPITATION BELOW THRESHOLD · B GROUP · BW DESERT · LESS DRY · BS STEPPE; H/K THERMAL SUFFIX

TEMPERATURE + RAIN SEASON → dryness threshold

PRECIPITATION BELOW THRESHOLD → B group → DRIER → BW desert → LESS DRY → BS steppe; h/k thermal suffix

CORE CLAIM
- TEMPERATURE + RAIN SEASON → dryness threshold
- PRECIPITATION BELOW THRESHOLD → B group

MECHANISM
- DRIER → BW desert
- LESS DRY → BS steppe; h/k thermal suffix

CONSEQUENCE / LIMIT
- TEMPERATURE + RAIN SEASON → dryness threshold
- PRECIPITATION BELOW THRESHOLD → B group

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LESS DRY → BS steppe; h/k thermal suffix.

04

STAGE 04 C-D THRESHOLD FIREWALL

C-D THRESHOLD FIREWALL · COLDEST MONTH · C/D DISCRIMINATOR · CONVENTION 1 · MINUS 3 C BOUNDARY · CONVENTION 2 · 0 C BOUNDARY · ANSWER RULE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
COLDEST MONTH → C/D discriminator	CONVENTION 2 → 0 C boundary	COLDEST MONTH → C/D discriminator
CONVENTION 1 → minus 3 C boundary	ANSWER RULE → state convention before mapping	CONVENTION 1 → minus 3 C boundary

CORE CLAIM
- COLDEST MONTH → C/D discriminator
- CONVENTION 1 → minus 3 C boundary

MECHANISM
- CONVENTION 2 → 0 C boundary
- ANSWER RULE → state convention before mapping

CONSEQUENCE / LIMIT
- COLDEST MONTH → C/D discriminator
- CONVENTION 1 → minus 3 C boundary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONVENTION 2 → 0 C boundary; ANSWER RULE → state convention before mapping.

05

STAGE 05 NORMAL-TO-TREND LADDER

NORMAL-TO-TREND LADDER · STATED 30-YEAR REFERENCE · OBSERVATION MINUS NORMAL · SPREAD AND RECURRENCE · MULTI-PERIOD DIRECTION WITH UNCERTAINTY

NORMAL → stated 30-year reference

ANOMALY → observation minus normal → VARIABILITY → spread and recurrence → TREND → multi-period direction with uncertainty

CORE CLAIM
- NORMAL → stated 30-year reference
- ANOMALY → observation minus normal

MECHANISM
- VARIABILITY → spread and recurrence
- TREND → multi-period direction with uncertainty

CONSEQUENCE / LIMIT
- NORMAL → stated 30-year reference
- ANOMALY → observation minus normal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VARIABILITY → spread and recurrence; TREND → multi-period direction with uncertainty.

06

STAGE 06 CLIMATE-BOUNDARY MODEL

CLIMATE-BOUNDARY MODEL · MEASURED VARIABLES · QUALITY CONTROL · COMPARABLE RECORDS · INTERPOLATION + THRESHOLDS · CLASS MAP · GROUND REALITY · TRANSITION, NOT WALL

STATIONS / GRIDS → measured variables → QUALITY CONTROL → comparable records → INTERPOLATION + THRESHOLDS → class map → GROUND REALITY → transition, not wall

• NORMAL → stated 30-year reference
• ANOMALY → observation minus normal

• VARIABILITY → spread and recurrence
• TREND → multi-period direction with uncertainty

• NORMAL → stated 30-year reference
• ANOMALY → observation minus normal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VARIABILITY → spread and recurrence; TREND → multi-period direction with uncertainty.

06

STAGE 06 CLIMATE-BOUNDARY MODEL

CLIMATE-BOUNDARY MODEL · MEASURED VARIABLES · QUALITY CONTROL · COMPARABLE RECORDS · INTERPOLATION + THRESHOLDS · CLASS MAP · GROUND REALITY · TRANSITION, NOT WALL

STATIONS / GRIDS → measured variables → QUALITY CONTROL → comparable records → INTERPOLATION + THRESHOLDS → class map → GROUND REALITY → transition, not wall

CORE CLAIM
- STATIONS / GRIDS → measured variables
- QUALITY CONTROL → comparable records

MECHANISM
- INTERPOLATION + THRESHOLDS → class map
- GROUND REALITY → transition, not wall

CONSEQUENCE / LIMIT
- STATIONS / GRIDS → measured variables
- QUALITY CONTROL → comparable records

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTERPOLATION + THRESHOLDS → class map; GROUND REALITY → transition, not wall.

07

STAGE 07 INDIA CONTROL MATRIX

INDIA CONTROL MATRIX · LATITUDE + ALTITUDE · THERMAL STRUCTURE · UPLIFT, RAIN SHADOW, COLD BARRIER · SEA DISTANCE · CONTINENTAL RANGE · MONSOON + WDS + OCEAN · SEASONAL VARIABILITY

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
LATITUDE + ALTITUDE → thermal structure	SEA DISTANCE → maritime / continental range	LATITUDE + ALTITUDE → thermal structure
RELIEF → uplift, rain shadow, cold barrier	MONSOON + WDs + OCEAN → seasonal variability	RELIEF → uplift, rain shadow, cold barrier

CORE CLAIM
- LATITUDE + ALTITUDE → thermal structure
- RELIEF → uplift, rain shadow, cold barrier

MECHANISM
- SEA DISTANCE → maritime / continental range
- MONSOON + WDs + OCEAN → seasonal variability

CONSEQUENCE / LIMIT
- LATITUDE + ALTITUDE → thermal structure
- RELIEF → uplift, rain shadow, cold barrier

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MONSOON + WDs + OCEAN → seasonal variability.

08

STAGE 08 INDIA CODE-MAP BOUNDARY

INDIA CODE-MAP BOUNDARY · COMMON TYPES · TROPICAL, DRY, SUBTROPICAL, HIGHLAND · EXACT CODE · DEPENDS ON DATASET AND THRESHOLD · DEPENDS ON SCALE AND INTERPOLATION · CITE SOURCE, PERIOD AND CONVENTION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
COMMON TYPES → tropical, dry, subtropical, highland	BOUNDARY → depends on scale and interpolation	COMMON TYPES → tropical, dry, subtropical, highland
EXACT CODE → depends on dataset and threshold	RULE → cite source, period and convention	EXACT CODE → depends on dataset and threshold

CORE CLAIM
- COMMON TYPES → tropical, dry, subtropical, highland
- EXACT CODE → depends on dataset and threshold

MECHANISM
- BOUNDARY → depends on scale and interpolation
- RULE → cite source, period and convention

CONSEQUENCE / LIMIT
- COMMON TYPES → tropical, dry, subtropical, highland
- EXACT CODE → depends on dataset and threshold

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXACT CODE → depends on dataset and threshold.

09

STAGE 09 CLASSIFICATION COMPARISON

CLASSIFICATION COMPARISON · TEMPERATURE + PRECIPITATION THRESHOLDS · MODIFIED THERMAL ZONATION · WATER BALANCE + PET · MATCH METHOD TO QUESTION PURPOSE · CLOSE DISTINCTION · A/B/C/D/E ARE MAJOR CLIMATE GROUPS, SECOND AND THIRD LETTERS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET	CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose	climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

CORE CLAIM
- KOPPEN → temperature + precipitation thresholds
- TREWARTHA → modified thermal zonation

MECHANISM
- THORNTHWAITE → water balance + PET
- CHOICE → match method to question purpose

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
- climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

09

STAGE 09 CLASSIFICATION COMPARISON

CLASSIFICATION COMPARISON | TEMPERATURE + PRECIPITATION THRESHOLDS | MODIFIED THERMAL ZONATION | WATER BALANCE + PET | MATCH METHOD TO QUESTION PURPOSE | CLOSE DISTINCTION | A/B/C/D/E ARE MAJOR CLIMATE GROUPS, SECOND AND THIRD LETTERS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
KOPPEN → temperature + precipitation thresholds	THORNTHWAITE → water balance + PET	CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
TREWARTHA → modified thermal zonation	CHOICE → match method to question purpose	climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

CORE CLAIM
- KOPPEN → temperature + precipitation thresholds
- TREWARTHA → modified thermal zonation

MECHANISM
- THORNTHWAITE → water balance + PET
- CHOICE → match method to question purpose

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and
- climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: A/B/C/D/E are major climate groups, second and third letters carry moisture/thermal meaning, and climate region differs from vegetation biome or agro-climatic zone even when boundaries broadly correlate.

10

STAGE 10 PLANNING OVERLAY

PLANNING OVERLAY | CLIMATE TYPE | LONG-PERIOD THERMAL-RAIN PATTERN | CROP AND SEASON CONTEXT | SOIL + LANDFORM + GROWING PERIOD | RISK LAYER | HAZARDS, EXPOSURE AND VULNERABILITY | EVIDENCE LIMIT

CLIMATE TYPE → long-period thermal-rain pattern

AGRO-CLIMATE → crop and season context → AGRO-ECOLOGY → soil + landform + growing period → RISK LAYER → hazards, exposure and vulnerability

CORE CLAIM
- CLIMATE TYPE → long-period thermal-rain pattern
- AGRO-CLIMATE → crop and season context

MECHANISM
- AGRO-ECOLOGY → soil + landform + growing period
- RISK LAYER → hazards, exposure and vulnerability

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Threshold classification simplifies transitional and highland mosaics
- map answers should state dataset/scale and avoid treating a code boundary as a permanent administrative or ecological line.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Threshold classification simplifies transitional and highland mosaics; map answers should state dataset/scale and avoid treating a code boundary as a permanent administrative or ecological line.

11

STAGE 11 CLIMATE ANSWER SPINE

CLIMATE ANSWER SPINE | PURPOSE, VARIABLES AND PERIOD | LETTERS AND THRESHOLDS | INDIA WITH TRANSITION CAVEAT | METHOD, USE, LIMITATION AND VERDICT

DEFINE → purpose, variables and period → DECODE → letters and thresholds → MAP → India with transition caveat → COMPARE → method, use, limitation and verdict

CORE CLAIM
- DEFINE → purpose, variables and period
- DECODE → letters and thresholds

MECHANISM
- MAP → India with transition caveat
- COMPARE → method, use, limitation and verdict

CONSEQUENCE / LIMIT
- DEFINE → purpose, variables and period
- DECODE → letters and thresholds

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MAP → India with transition caveat; COMPARE → method, use, limitation and verdict.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | A CLIMATES IN INDIA INCLUDE TROPICAL HUMID CLIMATES; AM | B CLIMATES ARE DRY | BSHW SEMI-ARID STEPPE AND BWHW HOT DESERT. | CWG IS THE GANGES/NORTH INDIAN PLAINS MONSOON CLIMATE WITH | H/E APPLIES TO THE HIMALAYA ABOVE HIGH ELEVATIONS/SNOW-LINE CONDITIONS. | TREWARTHA MODIFIES KOPPEN AND INCLUDES HIGHLAND CLIMATE FOR INDIA. | THORNTHWAITE USES PRECIPITATION–EVAPOTRANSPIRATION MOISTURE BALANCE, NOT TEMPERATURE ALONE.

OPTIONAL SOURCE DEPTH
- A climates in India include tropical humid climates; Am relates to heavy monsoon rain, while Aw marks savanna/dry-winter interior.
- B climates are dry: BShw semi-arid steppe and BWhw hot desert.
- Cwg is the Ganges/north Indian plains monsoon climate with dry winter.
- H/E applies to the Himalaya above high elevations/snow-line conditions.

ADVANCED DEBATE
- Trewartha modifies Koppen and includes highland climate for India.
- Thornthwaite uses precipitation–evapotranspiration moisture balance, not temperature alone.
- India has Koppen A, B, C and H in standard applied schemes.
- Am/Amw = West Coast and very wet monsoon belt; Aw = peninsular savanna interior.

LEGACY / NUANCE
- As = Tamil Nadu/Coromandel dry-summer tropical type linked to NE monsoon rainfall.
- BWhw = hot desert of western Rajasthan/Thar/Kachchh.
- BShw = semi-arid steppe/rain-shadow and NW margins.
- Cwg = Ganges type: north Indian plains, monsoon with dry winter.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels